

Barkın SARIKARTAL

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Portfolio: barkinsarikartal.github.io

Education

Çankaya University

Sep 2021 – June 2026

B.Sc. in Computer Engineering (100% Scholarship)

CGPA: 3.04

Experience

Çankaya University

Mar 2024 – June 2026

IT Department – Part-Time Student

- Worked as a part-time developer in the IT Department, contributed to university-level system development.
- Collaborated with a teammate to design and implement real-world embedded and software systems.
 - **Fleet Tracking System:** Developed a live tracking system for the Çankaya University's bus fleet. The system is built using an ESP32 microcontroller and a SIM7600 4G module for communication, and interacts with the university's servers via a REST API; assigned by the Board of Trustees of the university.
 - **Handheld RFID Card Reader:** Developed a Wi-Fi-enabled handheld RFID validator terminal for security guards at the university's main gate. In operation since January 2025; assigned by the Board of Trustees of the university.

ASELSAN

Jun 2025 – Jul 2025

Intern

- Worked on wireless communication protocols such as Wi-Fi, Bluetooth Classic and ESP-NOW with the ESP32 microcontroller and created fundamental test applications.
- Test applications: <https://github.com/barkinsarikartal/ESP-IDF-Projects>

Deneyap Türkiye

Sep 2023 – Dec 2023

Instructor of Electronic Programming and IoT

- Following a 14-week training program for middle and high school students, a project exhibition was held showcasing projects where students applied what they had learned.

Projects

Revumi (App Store & Play Store)

Aug 2026 – Sep 2026

Built a multiplayer game with Vue 3, TypeScript, FastAPI, and PostgreSQL, featuring WebSocket synchronization and reconnection handling. Directed AI-assisted development from planning and visual design through store release, reviewing generated code and validating changes with unit tests and CI checks. Personally configured Cloudflare and Railway infrastructure and managed deployment workflows.

Shyftel (App Store & Play Store)

Sep 2026 – Sep 2026

Built a cross-platform survival game using TypeScript, Three.js, Capacitor, and PostgreSQL, with server-side replay verification. Led AI-assisted coding and artwork creation through mobile store releases, reviewing implementation decisions and maintaining automated tests and CI checks. Personally configured Cloudflare and Railway infrastructure and managed deployment workflows.

Helma AI – AI-Powered Elderly Care Assistant (helma-ai.com)

Oct 2025 – June 2026

Developed a multimodal AI-powered elderly care application using Flutter, FastAPI, and MongoDB, with emotion recognition from video, audio, and text. Deployed and maintained the backend on AWS EC2 running Ubuntu 22.04.

ESP32 Based Retro Camera (github/barkinsarikartal/XIAO-ESP32S3-Sense_Retro_Camera)

Feb 2026 – May 2026

Built an ESP32-S3 camera using C++ and FreeRTOS, with photo capture, video recording with audio, and a live TFT viewfinder.

Skills

Programming: C, C++, Python, TypeScript, Java, C#, JavaScript, SQL

Web & Mobile: Vue 3, Flutter, FastAPI, Capacitor, REST APIs, WebSockets

Database & Cloud: PostgreSQL, AWS EC2, Cloudflare, Railway

Embedded Systems: ESP32, ESP-IDF, FreeRTOS

Tools & Platforms: Git, Linux/Bash, VS Code, Codex, Claude Code, GitHub Actions

Additional

Languages: Turkish (native), English (C1)

Exam Results: ALES - 89 (May 2026), YDS - 75 (November 2025), YKS - Ranking: 22.138/1.627.539 (June 2021)

Personal Traits: Compatible with teamwork, likes to follow technological developments, analytical thinker, fast learner